Response to the second referee report

Gregory Morse Peter Rakyta

7 November 2026

We have reorganized the manuscript as a theory paper, extended the analysis to the remaining involution strata, corrected native constructions, and made the remaining atlas gaps visible. The complete compiler evaluation is preserved in a standalone supplement. The response to the first report is retained separately as `first-referee-response.pdf`.

1. SYC, square-root-SWAP, and TK2 columns

We agree that a verified long word alone is insufficient for a reliable reference when a much shorter implementation is known.

The previous SYC fallback used six pulses for CNOT. We derived and checked an exact two-SYC controlled-phase chart with explicit local angles, all real angle branches, and literal scalar closure. It supplies the native controlled-block constructor and the CNOT fallback. The supplement gives the algebra and time-ordered circuit. The identity is accepted by a compositional polynomial certificate, not by a numerical fit. Cirq's prior analytical native decompositions are now cited; the two-pulse count is not claimed as new. DCX also has a checked literal one-TK2 word.

We have not completed exact short charts for the entire SYC and square-root-SWAP columns. Instead, as the report recommends, the atlas now annotates every remaining count gap against the independently recorded numerical witness. The separate audit covers 75 nonlocal two-qubit entries per column, finding verified witnesses for all 75 SYC entries and 24 square-root-SWAP entries. The other 51 searches are explicitly inconclusive. All 99 witnesses were independently replayed using the catalog matrices and the actual local alphabets; the largest literal entry error is about 1.1×10^{-11}. No GP gate is used. For a symbolic family a witness is one named parameter sample and is labelled accordingly, not a uniform family bound or an exact symbolic identity. The operations, substitutions, errors, failures, and versions are included in the package.

The unrestricted TK2 one-macro Cartan cap is also displayed on every remaining two-qubit TK2 cell that exceeds it. Literal determinant obstructions remain separate. Thus the one-DCX correction does not hide other unfinished local-frame constructions. The CSV `native_column_gaps.csv` records all audited cells and phase modes. The displayed atlas words remain exact upper bounds; none of these numerical fits or capacity annotations is relabelled as an exact word.

2. Motivation and extension beyond trace zero

The introduction now motivates controlled observables through Hadamard tests, controlled Pauli products, and reflection/oracle use in qubitization. The claim concerns exact small-block interaction cost, without asserting a measured algorithmic speedup.

A new proposition covers traces ± 2 and ± 4. Every trace-± 2 involution is locally conjugate to $\pm R_\theta$, where

$$R_\theta = I - 2|\psi_\theta\rangle\langle\psi_\theta|, \qquad |\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle, \quad 0 \le \theta \le \pi/4.$$

Explicit words give eight CNOTs or six unrestricted two-qubit gates across the family. Exact realignment gives

$$\det \mathcal{R}(R_\theta) = -\sin^2(2\theta)\,[2 - \sin^2(2\theta)].$$

Therefore the four-Ising theorem gives the interior CNOT lower bound four, and the general two-gate exclusion gives unrestricted lower bound three. Both upper words are checked with the Schmidt angle free.

At the product endpoint the controlled gate is locally CCZ, with minima six CNOTs and five unrestricted gates. At the maximally entangled endpoint it is locally Fredkin, with unrestricted minimum five and CNOT bounds five to seven. The sign of the reflection adds only a local control Z. Trace ± 4 gives zero interactions. This supplies normal forms and bounds for every spectral stratum, while explicitly leaving the trace-± 2 interior optimum open. It is not presented as a complete extension of the balanced cost classification.

3. SQiSW positioning

The invariant result is now a proposition. We cite Peterson–Crooks–Smith's monodromy-polytope treatment (including its corrected version) and identify Shende–Bullock–Markov's trace-invariant tests as the closest analogue. Huang et al. retain credit for the known two-pulse chamber region.

The main paper now gives the square-root-free version. With $a = \Re g_1$, $b = \Im g_1$, and $W = g_2 + 1$, membership is equivalent to

$$W \ge 0, \quad W^2 - 8a \ge 0, \quad W^4 - 16aW^2 - 64b^2 \ge 0.$$

For unitary inputs, $1/\det U = \overline{\det U}$, so these are real polynomial inequalities on matrix entries. The purpose is symbolic-family reasoning. We explicitly acknowledge that numerical canonicalization is already a small eigenvalue problem.

4. Theory-paper organization

The main paper now has one short software-and-data section. All width- and scope-resolved tables, SDK/Quartz/QUESO comparisons, OSR and chart ablations, reuse timings, scalar-program measurements, and the negative peephole experiment are in the standalone supplement. The numerical application's better local count and depth remain reported there. No negative result was removed from the data or replaced by an advantage claim. The full atlases, editable sources, exact checks, and numerical witnesses are bundled.

Minor points

We cite Yu–Ying's published Physical Review A 91, 032302 (2015) version; the controlled-product statement is its Proposition 2. The edge table now explicitly sets $t = \beta$ on the two fixed-α edges and $t = \alpha$ on the two fixed-β edges. It states $V_0 = K'QK'^\dagger$ and time order $K'^\dagger, C(Q), K'$. The balanced-cost table identifies its lower-bound dependencies: the four-Ising theorem at rank four and the unrestricted two-gate exclusion at rank two.

The C4X discussion no longer describes 36 CX as best known. It reports Quartz's verified 30-CX output and checks recent Huang–Palsberg and Vale no-helper-qubit routes through their recorded SDK implementations. Their measured C4X counts do not beat 30; this bounded check is not an exhaustive literature optimum. Full output circuits and literal replays are archived.

The author line names Gregory Morse and Peter Rakyta. Affiliations and the individual contribution allocation require author-supplied information; we have removed placeholder prose from the scientific manuscript and track these outstanding editorial fields separately. They must be settled before an external submission. No submission has been performed.